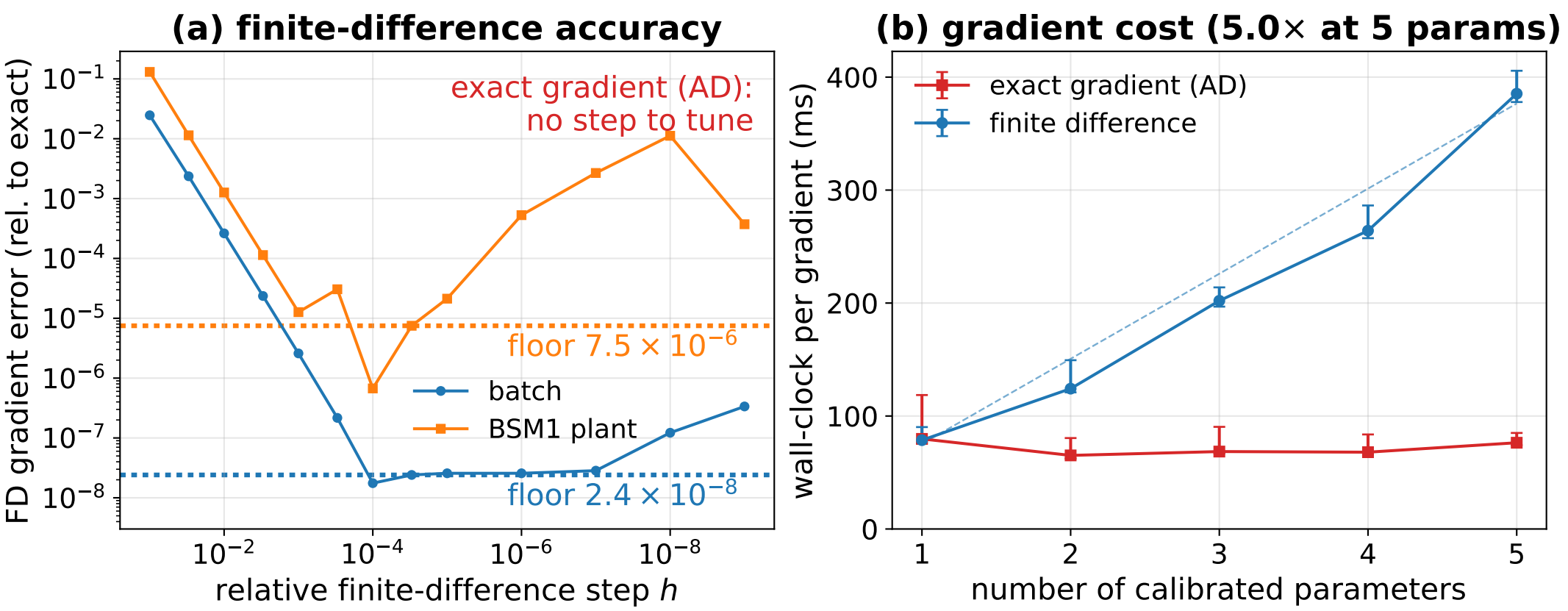
(a) finite-difference accuracy
FD gradient error (rel. to exact)
relative finite-difference step $h$
exact gradient (AD):
no step to tune
floor $7.5 \times 10^{-6}$
floor $2.4 \times 10^{-8}$
batch
BSM1 plant
(b) gradient cost (5.0× at 5 params)
wall-clock per gradient (ms)
number of calibrated parameters
exact gradient (AD)
finite difference